

BANARAS HINDU UNIVERSITY

B.A. (Hons.) — Semester I Examination, 2022-23

Paper No. MTB-102: Geometry

Time: 3 Hours

Maximum Marks: 70

Ref. No.: 057058

*Instructions: Answer **any five** questions including Question 1 which is compulsory. Figures in right-hand margin indicate marks.*

Question 1. (Compulsory — 2 marks each.)

- (a) Define (i) focus, (ii) directrix, (iii) eccentricity, (iv) latus rectum of a conic section.
- (b) Show that the general equation of a straight line in polar coordinates is $p = r \cos(\alpha - \theta)$, where (p, α) is the foot of the perpendicular from the origin.
- (c) Transform $ax + by + cz + d = 0$ into normal form $\ell x + my + nz = p$.
- (d) Define the angle between two planes.
- (e) Find the image of $(1, 3, 4)$ in the plane $2x - y + z + 3 = 0$.
- (f) Define shortest distance between two lines.
- (g) Show that a plane section of a sphere is a circle.
- (h) Find the sphere through $x^2 + y^2 = a^2$, $z = 0$ and the point (a, b, γ) .
- (i) Define reciprocal cones.
- (j) Define a right circular cylinder.
- (k) Obtain the tangent plane to $ax^2 + by^2 = 2cz$ at (α, β, γ) .

Question 2.

- (a) Show that the polar equation of a parabola is $\ell/r = 1 + \cos \theta$. [6]
- (b) If PSQ and $PS'R$ are two chords of an ellipse through foci S and S' , prove $\frac{1}{SP} + \frac{1}{SQ} \cdot \frac{1}{S'P} + \frac{1}{S'R}$ is independent of the position of P . [6]

Question 3.

- (a) Derive the equation of a plane in (i) normal form $(\cos \alpha)x + (\cos \beta)y + (\cos \gamma)z = p$ and (ii) intercept form. [6]

- (b) Find the plane through $(2, 1, -1)$ orthogonal to $x - y + z = 1$ and $3x + 4y - 2z = 0$. [6]

Question 4.

- (a) Derive the condition for two lines to be coplanar. [6]
- (b) Show the plane containing $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$, $y = 0$ parallel to $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$, $x = 0$ is $\frac{x}{a} - \frac{y}{b} - \frac{z}{c} + 1 = 0$ and find the shortest distance between them. [6]

Question 5.

- (a) A plane through (a, b, c) cuts the axes in A, B, C . Show the locus of the centre of the sphere through O, A, B, C is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 2$. [6]
- (b) If two spheres of radii r_1 and r_2 cut orthogonally, prove the radius of the common circle is $\frac{r_1 r_2}{\sqrt{r_1^2 + r_2^2}}$. [6]

Question 6.

- (a) Find the cone with vertex at origin and base $x = a$, $y^2 + z^2 = b^2$. Show the section by a plane parallel to XOY is a hyperbola. [6]
- (b) Find the cylinder with generators parallel to z -axis and guiding curve $x^2 + y^2 = z$, $x + y + z = 1$. [6]

Question 7.

- (a) Show the locus of intersection of three mutually perpendicular tangent planes to $ax^2 + by^2 + cz^2 = 1$ is $a^{-1} + b^{-1} + c^{-1} = x^2 + y^2 + z^2$. [6]
- (b) Prove that through any given point five normals can be drawn to a paraboloid. [6]